Exploratory Data Analysis (EDA) Summary Report

Geldium Delinquency Prediction Dataset / Task 1

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1 Introduction

The purpose of this report is to assess the completeness, consistency, and initial risk signals in Geldium's customer dataset before predictive modeling. The analysis reviews missing values, possible anomalies, variable definitions, and early relationships with the `Delinquent_Account` target. The dataset guide describes delinquency as a binary field (0 = No, 1 = Yes) and monthly payment history as On-time, Late, or Missed.

2 Dataset Overview

- The workbook contains 500 customer records and 19 columns.
- Key fields include `Age`, `Income`, `Credit_Score`, `Credit_Utilization`, `Missed_Payments`, `Delinquent_Account`, `Loan_Balance`, `Debt_to_Income_Ratio`, `Employment_Status`, `Account_Tenure`, `Credit_Card_Type`, `Location`, and `Month_1` to `Month_6`.
- Data types include numerical fields, categorical fields, a unique customer identifier, and a binary target.
- No fully duplicated rows were found; `Customer_ID` is unique across all 500 records.
- 80 records (16.0%) have `Delinquent_Account` = 1; the target is imbalanced (84% are 0, 16% are 1).
- `Employment_Status` contains inconsistent labels/capitalization: Unemployed (93), retired (87), Employed (82), EMP (81), Self-employed (80), employed (77). For example, 'Employed', 'employed', and 'EMP' may represent the same category and should be confirmed and standardized.
- `Credit_Utilization` ranges from 0.050 to 1.026. 4 records exceed 1.0 (100% if stored as a fraction). The guide states a 0–100% range, so these values and the fraction-versus-percent convention should be checked.
- `Credit_Score` ranges from 301 to 847, within the guide's typical 300–850 range.

3 Missing Data Analysis

These are proposed treatments, not changes applied to the source workbook. Before imputation, investigate why values are missing and whether missingness relates to delinquency or other customer attributes. Do not generate synthetic values simply to fill gaps; use synthetic data only with a clear purpose and validation.

4 Key Findings and Risk Indicators

- Observed target association is weak for the numerical fields in this initial check. The strongest absolute Pearson correlations with `Delinquent_Account` are still small: `Income`

Variable	Missing values	Proposed handling	Justification
`Income`	39 (7.8%)	Median imputation after checking distribution and missingness; preserve a missingness flag if useful.	Income is important and may be skewed by high values; the median is less sensitive to outliers than the mean.
`Loan_Balance`	29 (5.8%)	Median imputation after checking distribution and missingness; preserve a missingness flag if useful.	Avoids dropping otherwise usable customer records and is robust to skew.
`Credit_Score`	2 (0.4%)	Review the two records; use median imputation only if deletion is not justified and the values are plausibly missing at random.	Only 0.4% is missing, so either carefully justified row deletion or robust imputation may be reasonable.

Table 1: Missing values and proposed treatments.

(+0.045); `Account_Tenure` (−0.040); `Credit_Score` (+0.035); `Debt_to_Income_Ratio` (+0.034). These values do not establish useful predictive relationships by themselves.

- `Missed_Payments` does not separate the target in this extract: the mean is 2.85 for delinquent accounts and 2.99 for non-delinquent accounts. This unexpected result should be investigated; check definitions, data generation, and consistency between the target and payment history.
- Payment history (`Month_1`–`Month_6`) is a plausible risk signal based on the dataset definition, but its relationship to `Delinquent_Account` should be measured rather than assumed. Monthly values should be validated against the target.
- Credit utilization and debt-to-income ratio are plausible affordability/risk indicators. Confirm whether they are stored as fractions (0–1) or percentages (0–100), and investigate values outside the documented range.
- Credit score, income, loan balance, employment status, and account tenure may add context, but the current dataset review does not confirm that any of these is a strong predictor.
- The target is imbalanced (16% positive). Later model evaluation should include precision, recall, F1, and ROC-AUC rather than relying on accuracy alone.
- Use sensitive demographic/proxy variables carefully and assess fairness; location or employment category differences should not be treated as causal explanations without further analysis.

5 AI & GenAI Usage

GenAI can assist with summarizing columns, proposing data-quality checks, and comparing possible missing-data treatments. Its suggestions must be checked against the actual workbook and the Dataset Description Guide. No AI-generated synthetic values were inserted into this dataset for this report.

Example prompts used / suitable for this analysis:

- "Summarize the dataset dimensions, data types, missing values, duplicates, and out-of-range values. Report counts and percentages from the workbook."

- "Suggest missing-value treatments for Income, Loan_Balance, and Credit_Score, explaining assumptions and risks."
- "Compare Delinquent_Account with Missed_Payments, monthly payment history, Credit_Utilization, and Debt_to_Income_Ratio. Report observed statistics and flag unexpected results without assuming causation."
- "Check Employment_Status for inconsistent labels and recommend a transparent standardization rule."

6 Conclusion & Next Steps

The dataset has 500 records, no duplicate rows, and three fields with missing values: `Income` (39), `Loan_Balance` (29), and `Credit_Score` (2). The main quality concerns are inconsistent employment-status labels, credit-utilization values above 100% if interpreted as a fraction, and weak/unexpected observed associations between the delinquency target and payment indicators. Before modeling, standardize category labels, confirm ratio units and ranges with the data owner, investigate target/payment-history consistency, document and validate missing-data treatment, and then perform deeper univariate and bivariate analysis. Keep an audit trail of all transformations and evaluate fairness and class imbalance during modeling.

Scope note: This is an initial EDA summary of the supplied workbook, not a trained predictive model. Correlations are descriptive and do not imply causation.